NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED **MATHEMATICS** **M.M. 69**

REVISION ASSINGMENT # 16 (APPLICATION OF DERIVATIVES-02) TIME : 60 MIN.

SECTION–I(i)

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. The set of values of 'a' for which the function $f(x) = (a+2)x^3 - 3ax^2 + 9ax - 1$ is decreasing for all real x, is -

(A) $a \le -3$ (B) $a \le -2$ (C) $a \ge -3$ (D) $a \ge 3$

2. Complete set of values of m, for which function $f(x) = 3 + mx + e^{-x}$ is always decreasing, is -

(A) $[0, \infty)$ (B) $(-\infty, 0]$ (C) $[2, 5]$ (D) $[7, 17]$

3. Which of the following curve has constant length of subnormal at every point -

(A) $x^2 = y$ (B) $y^2 = x$ (C) $x^2 + y^2 = 1$ (D) $x^2 - y^2 = 1$

4. Let f, g are two continuous and twice derivable functions such that $f(0) = f(3) = 0$; $f(1).f(2) < 0$; $g(0) = g(3) = 0$. Number of roots of equation $f''(x).g(x) + f'(x).g'(x) = 0$ in $(-1,5)$ cannot be

(A) 2 (B) 3 (C) 4 (D) 5

5. Given $f(x) = \begin{cases} |x-1|+5 & \text{if } x \neq 1 \\ 8 & \text{if } x = 1 \end{cases}$, then at x = 1, $f(x)$ has-

(A) a local maximum (B) a local minimum

(C) neither maximum nor minimum (D) a unique tangent with finite slope.

Rankers Academy JEE

6. Function $f(x) = \dfrac{x^2 - 2}{\sqrt{1+x^2}}$

(A) is always increasing (B) is always decreasing

(C) has exactly one point of minima (D) has exactly one point of maxima

7. If $f(x) = \begin{cases} 2x+3 & x < 0 \\ 5 & x = 0 \\ x^2+7 & x > 0 \end{cases}$, then at x = 0, $f(x)$

(A) is decreasing (B) is increasing

(C) is neither increasing nor decreasing (D) has local maxima

8. Let $f(x) = \int_0^{x^2} (t-1)(t-4)(t-9)dt$, then

(A) $f''(x) = 0$ have 4 distinct positive solutions (B) $f'''(x) = 0$ have 2 distinct positive solutions

(C) $f'''(x) = 0$ have 3 distinct positive solutions (D) $f(x)$ have 6 critical points.

Space for Rough Work

For More Material Join: @JEEAdvanced_2024

9. Number of non negative integers which satisfy the inequation $f(f(f(x))) > f(f(-x))$, where $f(x) = 30 - 2x - x^3$ is-

(A) 0 (B) 1 (C) 2 (D) 3

10. A box is to be made with square base and open top. If the area of material used is 48 sq. meter, then maximum volume of the box is-

(A) $48m^3$ (B) $16m^3$ (C) $32m^3$ (D) $36m^3$

SECTION–I(ii)

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

11. Let the parabola $y = x^2 + ax + b$ and $y = x(c - x)$ touches each other at point (1,0), then-

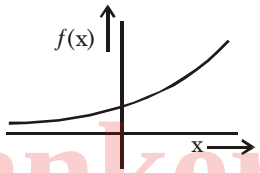
(A) $a + c = -2$ (B) $b + c = 3$ (C) $a = -3$ (D) $b = 2$

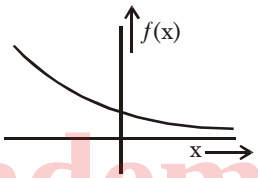
12. Let $f(x) = \begin{cases} (x-2)^2 + 3 & x \geq 2 \\ k + x^2 & x < 2 \end{cases}$, then

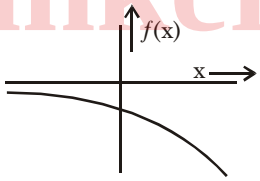
(A) $f(x)$ have local minima at $x = 2\ \forall\ k \geq -1$ (B) Range of $f(x)$ is $[3, \infty)\ \forall\ k \in [3, \infty)$

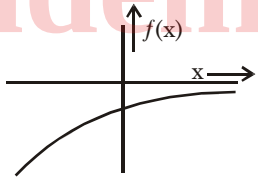
(C) $f(x)$ is non-differentiable $\forall\ k \in R$ (D) $f(x)$ is continuous $\forall\ k \in R$

13. Let $f(x)f'(x)f''(x) > 0\ \forall\ x \in R$, then which of the following is(are) best possible graph of $f(x)$

(A) 

(B) 

(C) 

(D) 

Rankers Academy JEE

14. Let a function $f : R \to R$ satisfies functional equation $f(x + y) = f(x).f(y)$ where $f(0) \neq 0$ and $f'(0) = 1$.

Given $g(x) = \dfrac{3x}{f(x)}$, then -

(A) The value of $g'(\ell nx)$ at $x = \sqrt{e}$ is $\dfrac{3}{2}(e)^{-\frac{1}{2}}$ (B) g(x) attain local minima at x = 1

(C) g(x) attain local maxima at x = 0 (D) if g(x) = k has two solutions then $k \in \left(0, \dfrac{3}{e}\right)$

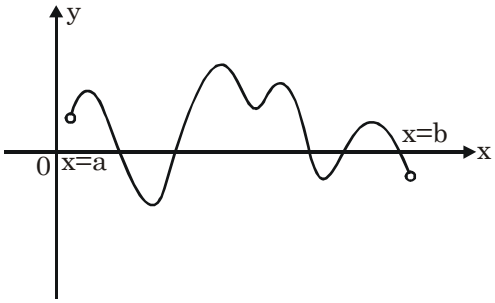
Space for Rough Work

For More Material Join: @JEEAdvanced_2024

SECTION–I(v)

Matching list type (4 × 4) (Single option correct) (3 Marks each, –1 for wrong answer)

15. Consider the curve for $y = f'(x)$ as shown below



then for the curve $y = f(x)$.

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		**List-II**
(P)	Number of points of maxima	(1)	7
(Q)	Number of points of minima	(2)	3
(R)	Number of inflection points	(3)	5
(S)	Number of stationary points	(4)	2

Codes :

	P	Q	R	S
(A)	3	4	1	2
(B)	4	1	2	3
(C)	2	1	4	3
(D)	2	4	1	3

Rankers Academy JEE

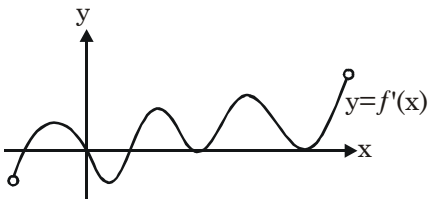
Space for Rough Work

For More Material Join: @JEEAdvanced_2024

SECTION–II (iii)

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. Let $f : R \to R$ and graph of function $y = f'(x)$ is shown below



Let a = number of point of inflection of $y = f(x)$

b = number of point of local minima of $f(x)$

c = number of point of local maxima of $f(x)$

d = number of critical point where neither maxima nor minima occur, then value $\frac{a-c}{a+b+d}$ is equal to

2. Let $f(x)=\frac{e^x}{x}$. If $f(x) = k$ has two distinct solution then smallest integral value of k is

3. Let $f(x)=\begin{cases} e^x - 2x, & 1 \le x \le 2 \\ 2\sin x + x, & 2 < x \le 2\pi \end{cases}$, then least value of $f(x)$ in $x \in [1, 2\pi]$ is equal to

4. The number of distinct real roots of equation $2x^4 - 8x^3 + 8x^2 - 1 = 0$ is n, then the value of $\frac{1}{n}$ is

5. If tangent to the curve $xy = x^2 + 1$ at (α,β) is normal to the curve $x^2 + y^2 + 2fy + c = 0$, then $\frac{1}{|f.\alpha|}$ is

Rankers Academy JEE

Space for Rough Work

REVISION ASSINGMENT # 15 APPLICATION OF DERIVATIVES-01 MATHEMATICS

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	B	C	B	C	A	D	C	C,D	B,C	B,C,D
	Q.	11	12	13	14						
	A.	B	D	B	D						
SECTION-III	Q.	1	2	3	4						
	A.	4	1	9	1						

For More Material Join: @JEEAdvanced_2024